

CHAPTER ONE

INTRODUCTION TO MATERIAL MANAGEMENT

1.1 What is materials management?

Materials management is the process of planning, organizing, implementing and controlling the tasks associated with the flow of material to through and out of an organization in an integrated fashion.

The main point of the definition includes the following: -

- Material management, as other management branch employees the basic function of management:
 - a) Planning
 - b) Organizing
 - c) Directing and
 - d) Controlling
- All these functions target at the optimum use and allocation of materials.
- Materials should be managed before, during and after the process.
Input → Process → output
- MM (Materials Management) provides an integrated system approach to the coordination of materials activities and control of total material costs. It advocates assigning to a single operating department all major activities which contributes to the cost of materials.

TYPES OF MATERIALS

Materials are any resource use in the direct conversion of a product or the aid to the conversion process.

Types of materials are usually classified based on accounting purpose and based on their use.

Materials classified based on accounting purpose includes:

- a) **Raw materials:** basic input, components and sub assemblies.
- b) **Work in process:** semi-processed, semi-finished material.
- c) **Finished goods:** fully completed material
- d) **Supplies**

Material classification based on their use includes:

- a) **Production Item:** materials that becomes part of produced item, they can be
 - i. Fabricated part or
 - ii. Component item
- b) **Capital item**
 - Are item that is used to produce the physical products.
 - Are supposed to use for any period of time.
 - Are classified as:
 - i. Installation items: major capital that require huge investment
e.g Machines, tools, computers, equipment
 - ii. Accessory: equipments- required relatively less investment
- c) **Operating items:** used to facilitate production and other operations
e.g maintenance, repair and operating supplies to be used as needed for operations and maintenance.

1.2 IMPORTANCE OF MATERIALS MANAGEMENT

Do you think that Material management is important only for controlling of material or what do you think materials management could be used for? Do you think the importance of material management just limited to the process of material in the organization?

In general, Materials Management (MM) is important for three reasons:

1. Customer Service
2. Cost reduction
3. Administrative efficiency

1. Customer Service

Materials management (MM) serves other units/departments of the organization. This service support the goal of fast process throughout the organization, and include:

- Uninterrupted flow of materials.
- Assistance to customer-repair, replants or delivery.
- Reduction in parts shortage.
- Average transit from warehouse to customer
- Minimum of transportation delays.
- Accurate inventory counts.
- Good forecast accuracy.

2. Cost reduction

The importance of material management can be realized when it is said that;

- Purchase account for nearly 50% of an organization's annual expenditures.
- Nearly 80% of working capital is finding up in the inventory.
- 5% saving of materials cost will substantially increase the profit margin of an organization.

Cost reduction effort related to profit and effective material management certainly make deference.

Specific cost reduction measures include:

- Reduction in material cost.
- Reduction in storage space.
- Reduction in physical inventory.
- Increase in inventory turnover.
- Reduction in transportation cost.
- Reduction in setup time.

3. Administrative efficiency

The last area is administrative efficiency which covers quality of measures related to department efficiency in operation and development of people.

The measure includes:

- Material management budget Vs actual
- Ratio of material budget to sales.
- Ratio of key people among functional areas.
- Installation of productivity-enhancing systems.

Scope of Material Management

Dear student, do you think the function of material management is similar to other management function?

The main functions that come under the umbrella of material management are:

1. materials demand forecasting: follows from sales for cost
2. purchasing acquisition of the kinds and quantities of materials required by department
3. inventory control: planning and maintains stocks of raw materials, tools, supplies and other materials, classification
4. Traffic/transportation: includes both incoming and outgoing transportation also called logistics management (physical distribution)
 - a) traffic control involves
 - the selection of carriers
 - documentation of shipments
 - study of carrier services and rates
 - evaluations of carrier performance
 - b) traffic concerned with assessing the total cost of transportation including:
 - loading and unloading

- methods of packaging
 - transit time
 - developing techniques for reducing overall transportation costs
5. Store and material handling involves:
- Receiving
 - In plant materials movement
 - Insurance
 - Documentation, etc.
6. Disposal of scrap and surplus materials

Organization for materials management

By organization for material management, we mean the frame work (structure) with in tasks and responsibilities are discharged.

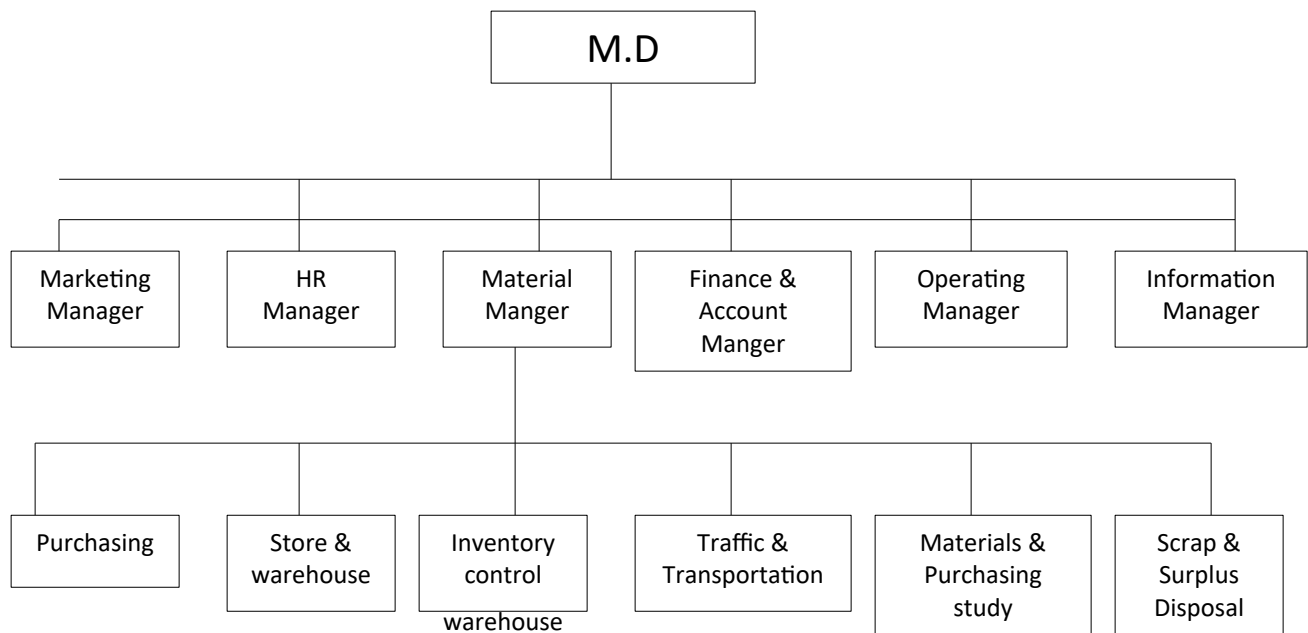
- It is question of where and how should material management be located in the organization structure
- Its position in organization structure depends on type and size of the organization

There are two approaches of organization for material management

Dear learner, you need to pay attention to the diagram below that describes the two approaches organization for material management.

1) Integrated structure

- This is a modern approach where all activities of materials headed by materials manager with other sub functions under him/her.
- It integrates of all materials related activities



As materials manager how your department is integrated to the structure or the function of your organization.

Benefits of the integrated approach:

- 1) equal status with other functional mangers
- 2) can view the total commitment on his/her department
 - produce a policy
 - establish procedures and budgets
- 3) Reports to higher level in the organization thus greater scope to influence board divisions affecting his area of operations and interfacing ones.

- 4) scope to evaluate the strengths and weakness throughout the white of the materials management department
- 5) better accountability: though centralization of authority and responsibility for all aspects of materials function, a clear cut responsibility established
- 6) better performance: as all inter-related functions are integrated organizationally, greater speed and accuracy, results in communication
- 7) better service

2. Differentiation /Defused approach

- this is a very old approach where individual material functions are assigned to other functional areas
- this organization assignment of individual functions to other major functions will based on
 - i) “most use” criterion

Example:

Purchasing might be assigned to the operations department since the major money is spent for raw materials

- Traffic/transportation: might be assigned to marketing department since he/she would be responsible for delivery of finished items for business
- Material and purchase study to R & D department
- ii) critical relationship criterion

Example: Inventory control might be assigned to finance in view of the dollar
Investment involved

- iii) executive interest basis

Example: value analysis function might be assigned to engineering department.

Since the engineering may be the one who has promoted and pushed this activity.

The point is that, the responsibility for different, but interrelated, functions becomes widely dispersed throughout the organization structure.

Benefits/Advantage

- 1) no additional level of mgt is created
- 2) no need for additional employees, and other resources
- 3) avoids additional horizontal communications

Disadvantages

- 1) overseeing of materials in each department
- 2) subordination of material function
- 3) Expenditure on materials is a higher percentage of a company's total annual expenditure. There would be no senior manager at an influential level to control it.
- 4) There would be no one capable of promoting effective to coordinated materials plan to meet materials needs of different sections of the organization.
- 5) Problem of coordination and communication
- 6) No job specifications
- 7) No employee development

Concepts and relations of materials management with purchasing, supply and logistics management

Purchasing: sub functions comprise the essential activities associated with the acquisition of materials used in the operations of an organization. The major types of activities are:

- Coordination with user departments to identify material needs
- Identification of potential suppliers
- Negotiation with potential suppliers
- Selection of suppliers
- Issuance of purchase order
- Market studies for important materials
- Analysis of proposal

- Administration of contract and resolution of related problems
- Maintenance of purchasing records

Supply management

Supply management is a process responsible for the development of forms total supply system both internal and external components. At operations level it includes and expands the activities of the purchasing function specific activities included in supply management are:

- conduct of a all purchasing functions
- early purchasing involvement (EPI) and early supplier involvement (ESI) in
 - product design and development
 - material specification
- heavy use or cross-function teams in supplier qualification and selections
- heavy use of purchasing partnering arrangement and strategic alliances with suppliers
- development of strategic acquisition plans for all materials
- active participation in the corporate strategic planning process

Generally, the major characteristic that differentiates it from purchasing is that **it focuses heavily on the strategic aspects of the key elements of the firm's supply system.**

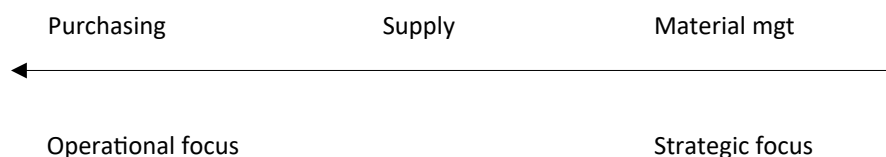
Materials Management: an organization concept designed to enhance coordination and control of the various material activities.

The specific material activities (functions) typically found in material managements are:

- all purchasing activities
- all supply activity
- inventory management
- materials handling
- traffic/transportation
- disposal management

Difference among the three functional

1. Purchasing and supply management are component of materials management
2. Materials management includes other material activities
3. Material management is largely a strategic focus



Relations of material management with other department of organizations

Broadly, the material management will have to work in close coordination with the available departments of the organization.

Operation, Marketing, Finance, HRM

1. M.M. and Operation

- Materials management department must work with the operations department and informed about the availability of raw materials, anticipated delays so that re-scheduling of production could be done and costly stock outs avoided.
- The expertise of the materials management could be used by the operation department. In purchasing capital materials.
- Operation will have to keep materials management department informed about its plans and schedules so that material requirements can be planned in advance.

1. M.M. and Finance:

The finance department also has to work in close coordination with material management department in:-

- anticipating the availability of funds
- payment of bills of purchases
- accurate record of purchases
- source of accounting documents

2. M.M. and Marketing

Marketing department will **give advance information on sales forecast** on which material forecast is based on materials management through efficient operation can keep product prices at a

Competitive level and thus help the marketing department in its operation.

3. M.M. and HRM

- HRM perform the basic recruitment selections promotion and other employees aspects of materials management department

MATERIAL DEMAND FORECASTING

2.1. MEANING, CHARACTERISTICS AND USES OF FORECASTING

Forecasts are estimates of the occurrence, timing or magnitude of future events. They give operation managers a rational base for budgeting, capacity planning, sales promotion and inventory, personnel and material management. Forecasting is the basis of planning ahead even though the actual demand is quite uncertain. Planning means, foresight plus actions to make things come out to our advantage. It involves estimation of the future, and of particular interest here is the expected demand of company's product. Therefore, forecast of future demand is the link between company's internal expectations with outside environment that permits planning function to commence activities. Forecast affects decisions and activities throughout the organization in Finance, HRM, marketing, operations and other functions.

CHARACTERISTICS OF FORECASTS

The following are characteristics of forecasts: -

1. Forecasting technique generally assumes the same underlying causal system that existed in the past will continue to exist in the future.
2. Forecasts are rarely perfect; actual results usually differ from predictive values.
3. Forecasts for a group of items tend to be more accurate than forecasts for individual item, because forecasting errors among items in a group usually are smaller than that of individual item.
4. Forecast accuracy decreases as the time period covered by the forecast –time horizon increases.

USES OF FORECAST

Components, subassemblies or/inquired services that are part of the finished product may not require formal forecasts (i.e. not all materials require formal forecast). Forecast should be used for end items and services that have uncertain demand. The purpose of forecasting activities is to make the best use of present information to guide decisions towards the objective of the organization in general. Accurate projections of future activity levels can minimize short term fluctuations in production and help balance workloads. These reduce hiring, firing and overtime activities and help maintain good labour relationship.

Good forecasts also help managers have appropriate level of materials available when needed. By anticipating employment materials needs, the forecasts enable managers to make better use of facilities and give improved service to customers.

Generally, good forecast:

- Could improve employee relation.
- Improve materials management.
- Helps to have better use of capital and facilities and
- Improves customers services

2.2 TYPES OF FORECASTING

There are two types of forecasting techniques. They are

1. Qualitative forecasting technique.
2. Quantitative forecasting technique.

2.2.1 QUALITATIVE FORECASTING TECHNIQUE

Qualitative forecasting technique is a technique that is used when there is no historical data available about past performance. These forecasting techniques are subjective and judgemental in nature and most of the time they are based on opinion and expertise judgment. Qualitative forecasting techniques rely on analysis of subjective input obtained from customers, sales person, managers and experts.

Types of qualitative forecasting techniques:

There are four common types of qualitative forecasting technique. They are:

1. Expert opinion method.

2. Sales opinion
3. Consumer surveys
4. Delphi technique

1. Expert opinion method

One of the most simple and widely used methods of forecasting which consists of collecting opinions and judgements of individuals who are expected to have the best knowledge of current activities or future plans. This technique has its own advantages and disadvantages.

Advantage

- ✓ Decision is fast
- ✓ Responsibility and accountability is clear
- ✓ Brings together the considerable knowledge, skill and talent of various managers
- ✓ Managers (experts) will acquire experience that is obtained in the decision

Disadvantage:

- ✚ Probably poor forecast (due to lack of experience).
- ✚ Domination by one or few manager.
- ✚ Diffusing responsibility for the forecast over the entire group may result in less pressure to produce a good forecast.

2. Sales force opinion

In this method, the sales representatives are required to estimate the demand for each product and the forecast of each sales representative is consolidated to prepare the overall forecast for the company.

This forecasting technique has also its own advantages and disadvantage.

Advantage:

- ✓ It can reset in quality forecast
- ✓ This pools together knowledge.
- ✓ Can see from different approaches.

Disadvantage:

- ✚ Time taking decision.
- ✚ Influenced by majority high status persons.
- ✚ Advocacy of responsibility.

3. Consumer survey:

This forecasting technique based on the data which is collected from the consumers. Because it is the consumers who ultimately determine demand, it seems important to solicit information from them.

Advantage:

- ✓ Tap information that may not be available elsewhere.
- ✓ Enhance the quality and accuracy of forecasts.

Disadvantage

- ✚ Experience and knowledge is constricting.
- ✚ Expensive and time consuming.

4. Delphi method:

This is a qualitative method forecasting which involves the development, distribution, collection and analysis of series of questioners to get the views of expertise that are located at different geographic areas to generate the forecast. A moderator compiles results and formulates a new questioner that is again submitted to the same group of experts. The goal is to achieve a consensus forecast.

Advantage

- ✓ The tendency of process loss is avoided/minimized.
- ✓ No influence of the majority.

Disadvantage

- ✚ It takes time to reach a consensus
- ✚ Coordination and interpretation difficult.

2.2.2 QUANTITATIVE FORECASTING TECHNIQUE

There are two types of quantitative forecasting technique:

1. Time series analysis.
2. Causal method.

1. Time series:

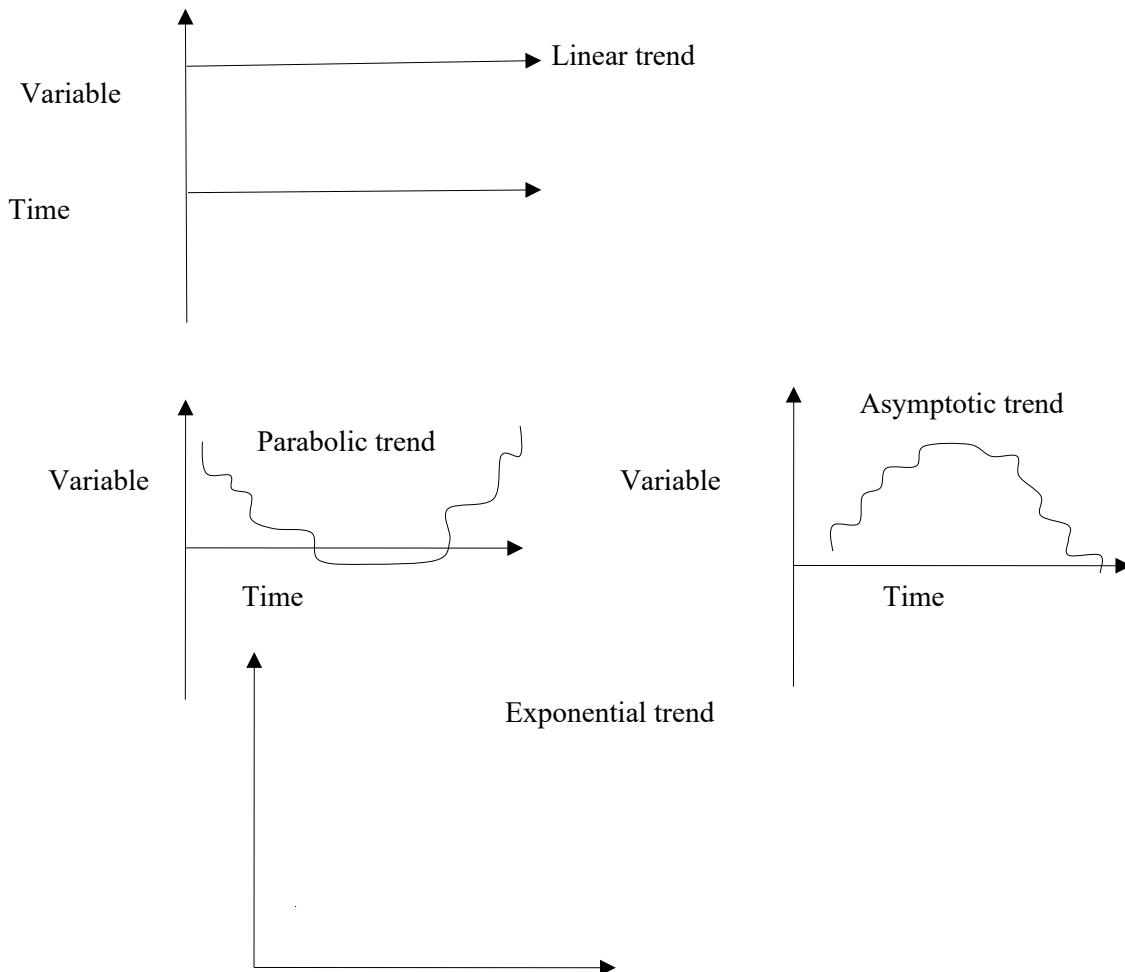
A time series is a set of some variable (demand) overtime (e.g. hourly, daily, weekly, quarterly annually).

Time series analyses are based on time and do not take specific account of outside or related factors.

Forecasting technique based on time series data are made on the assumption that history follows a pattern that will continue. Time series analysis is a time-ordered series of values of some variables. The variables value in any specific time period is a function of four factors.

- a) Trend
- b) seasonality
- c) cyclic
- d) Randomness

a) Trend: - is a general pattern of change overtime. It represents a long time secular movement, characteristic of many economics service.



b) Seasonality-refers to any regular pattern recurring with a time period of no more than one year. These effects are often related to season of the year.

- Seasonality is regularly repeating upward or downward movements in series value that can be tied to recurring events.

Example

- Weather variations-sales of winter and summer.
- Vacations or holidays- airline travel, greeting card, visitors at tourists and resort centres.
- Theatres demand on weekends.
- Daily variations: banks may over crowded during the afternoon

Seasonality in time series is expressed in terms of the amount that actual sales deviate from the average value of a series. There are two models of seasonality: additive and multiplicative.

Additive Model:

In the additive model, seasonality is expressed as a quantity (e.g 20 units) which is added or subtracted from the series average in order to incorporate seasonality.

Multiplicative model:

In the multiplicative model, seasonality is expressed as a percentage of the average (or trend) amount (e.g 1.10) which is then used to multiply the value of a series to incorporate seasonality.

- c) **Cycle-** are long-term swings about the trend line and are usually associated with a business cycle (phases of growth and decline in business cycle)
- d) **Randomness** – are sporadic effects due to chance and unusually occurrence

TYPES OF TIME SERIES ANALYSIS

A) Simple moving average

A simple moving average is obtained by summing and averaging values from a given number of periods repetitively, each time deleting the oldest value and adding the new value.

$$SMA = F_t = \frac{A_{t-1} + A_{t-2} + A_{t-3} + \dots + A_{t-n}}{n}$$

$$= \frac{\sum_{i=1}^n A_{t-i}}{n}$$

Where:

SMA= simple moving average

F_t = Forecast for period t

A_{t-i} = Actual demand in period t-i

n = number of periods (data points) in the moving average

Simple moving average is preferable if the demand for a product is neither growing nor declining rapidly and also does not have any seasonal characteristics

EXAMPLE 1

A food processor uses a moving average to forecast next month's demand. Past actual demand (in unites) is shown in the following table.

Months	1	2	3	4	5	6	7	8
Actual demand	105	106	110	110	114	121	130	128

Required:

- a) Compute a simple 5 month moving average to forecast demand for month 9.
- b) Find simple 5 month moving average to forecast the demand for month 10 if the actual demand for month 9 is 123.

Note: in moving average, as each new actual value becomes available, the forecast is updated by adding the newest value and dropping the oldest value and computing the average. Consequently the “forecast” moves by reflecting only the most recent values.

Advantage:

- ✓ Easy of computation.
- ✓ Easy of understanding

Disadvantage:

- All values in the average are weighted equally. The oldest value has the same weight as the most recent value.

Example 2

Sunrise bakery makes cakes and supply to the market. The following data shows their daily demand for the last four weeks. The bakery is closed on Saturday. So Friday's production must satisfy demand for Saturday and Sunday.

	4weeks ago	3weeks ago	2weeks ago	Last week
Monday	200	400	300	400
Tuesday	200	100	100	300
Wednesday	300	400	300	500
Thursday	800	900	400	100
Friday	500	200	300	400
Saturday	<u>800</u>	<u>700</u>	<u>600</u>	<u>500</u>
Sunday				
Total	2800	2700	2000	2200

Required: make forecast for this week on the following bases.

- Daily, using a simple four week moving average?
- Daily, using a simple three week moving average?
- Weekly, using a simple four week moving average?
- Weekly, using a simple three week moving average?

B) Weighted moving average

In weighted moving average, the weight is given in such a way that more weight is given to the most recent value in the time series. Weights can be percentages or any real numbers.

In weighted moving average, forecasts are calculated by:

$$F_t = WMA = W_1A_{t-1} + W_2A_{t-2} + \dots + W_nA_{t-n}$$

$$= \sum_{i=1}^n W_i A_{t-i}$$

Where;

F_t = forecast in time t

WMA = weighted moving average

W = weight

A = Actual demand value

Example 1

A department store may find that in a four month period the best forecast is derived by using 40% of the actual demand for the most recent month, 30% two months ago, 20% three months ago and 10% of four months ago. The actual demand were as follows

Month	1	2	3	4
Demand	100	90	105	95

Required: Compute weighted four-month MA for month five?

Example 2

See example 2 above in simple moving average and Make a daily base forecast for this week using a weighted average of 0.40 for last week, 0.30 for two weeks ago, and 0.20 for three weeks ago and 0.10 four weeks ago.

C) Simple Exponential smoothing

The other method of time series forecasting method is simple exponential smoothing which weights past data in an exponential manner; so that most recent data carry more weight in the moving average.

With simple exponential smoothing, the forecast is made up of the last period forecast plus a portion of the difference between the last period actual demand and the last period actual demand and the last period forecast.

Mathematically:

$$F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$$

Where:

F_t = forecast for period t .

F_{t-1} = forecast for the previous period.

α = smoothing constant ($0 < \alpha < 1$).

A_{t-1} = actual demand for the previous period.

The difference between the actual demand and the previous forecast (i.e. $A_{t-1} - F_{t-1}$) represents the forecast error.

As we observe from the equation, each forecast is simply the previous facts plus some correction for demand in the last period. If actual demand was above the last period forecast, the correction will be positive, and if the actual demand was below the last period forecast, the correction will be negative.

Exponential smoothing is the most widely used of all forecasting techniques, because

1. Exponential forecasting model provides closer forecasts to actual demand.
2. Formulating an exponential smoothing model is relatively easy.
3. The user can easily understand the model.
4. It requires little computation
5. It requires only three pieces of data
 - The most recent forecast.
 - The actual demand of the previous period.
 - The smoothing constant.

Example 1

The production supervisor at a fiber board plant uses a simple exponential smoothing technique ($\alpha=0.2$) to forecast demand. In April, the forecast for 20 shipments, and the actual demand was for 20 shipments. The actual in May and Jun was 25 and 26 shipments. Forecast the value for June.

Example 2

Demands during the past six months have been as follows:

January	115
February	123
March	132
April	134
May	140
June	147

Using simple exponential smoothing with $\alpha=0.70$, if the forecast for January had been 110, compute the exponential smoothed for each month through July.

2. Casual forecasting method:

The second type of quantitative forecasting technique is casual forecasting method. Casual forecasting techniques rely on identification of related variables that can be used to predict values of the variables of interest (demand). Casual methods are used when historical data are available and the relationship between the factor to be forecasted.

Example

1. Sales of beef may be related to
 - Price of beef
 - Price of substitutes, such as chicken, pork and lamb

2. Real state price are usually related to
 - Property location
 - Square footage
3. Crop yield are related to
 - Soil condition
 - Amount and timing of water
 - Fertilizer application

Regression and correlation method

Regression and correlation techniques are means of describing the association between two or more variables. More specifically, regression and correlation methods are related to the following issues.

- i. Bringing out the nature of relationship between any two variables, say X and Y.
- ii. Measuring the rate of change in one (the dependent) variables associated with a given change in the other (independent) variable.
- iii. Evaluating the strength of the relationship and quantifying the closeness of such relationship.

Regression

It is concerned about the first two issues:

- Bringing out the nature of relationship between any two variables, say X and Y.
- Measuring the rate of change in one (the dependent) variables associated with a given change in the other (independent) variable.

Regression means 'dependence' and involves estimating the value of the dependent variable Y, from an independent variable X.

They are two types of regression:

1. Simple Regression.
2. Multiple Regression.

In *simple Regression* only one independent variable is used, and the general form of this simple regression is $Y=b+aX$, where as in multiple regression two or more independent variables are involved, and multiple regression take the general form of :

$$Y=b+aX_1+cX_2+\dots+zX_n$$

Correlation

Correlation is concerned about evaluating the strength of the relationship and quantifying the closeness of such a relationship.

Simple Linear Regression and Correlation.

In simple linear regression, only one independent variable is used and the model takes the form:

$$Y=b+aX$$

Where:

Y= predicted (Dependent) variable, demand

a= value Y at X=0

b= slope of the line.

Note: -

1. It is convenient to represent the value of the predicted variable on the Y-axis and value of the predictor variable on the X-axis.
2. The coefficients **a** and **b** of the line are obtained by using the formula.

$$b = \frac{n \sum x_t y_t - \sum x \sum y}{n \sum x_t^2 - (\sum X_t)^2}$$

$$a = \frac{\sum y_t - b \sum X}{n}, \text{ or } a = \bar{y} - b \bar{x}$$

Where

n= number of period observation

Example

The general manager of a building materials production plant feels the demand for plaster board may be related to the number of construction permits issued in the country during the previous quarter. The manager has collected the data shown in the accompanying table.

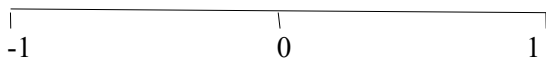
Construction permits	Plaster board shipments
15	6
9	4
40	16
20	6
25	13
25	9
15	10
35	16

Required

- Drive a regression forecasting technique?
- Determine plaster board demand when the number of shipment is
 - 30
 - 35
 - 40
- Compute coefficient of determination (r^2) and coefficient of correlation

Note: - Coefficient of correlation(r) =
$$\frac{n \sum Xy - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

Correlation coefficient, r is a number between -1 and 1



Correlation coefficient can be positive, Zero or Negative.

Positive $r \Rightarrow Y$ increases with increase in X .

Negative $r \Rightarrow$ Decrease with increase in X .

$r=1$ perfect positive relation

$r=0$ lack of any relationship between the two variables.

$r=-1$ perfect negative relationship.

Coefficient of correlation, r overstates the degree of relationship. It is also devoid of a meaning as explicit as that of the coefficient of determination,

Coefficient of determination r^2 ranges from 0 and 1, and it is a more objective and definitive measure of the degree of relationship.

2.3 Accuracy and control of Forecast.

Forecasting accuracy relates to the magnitude of forecasting error.

Forecasting Error at time t = Actual demand at time t - Forecasted demand at time t
 $e_t = A_t - F_t$

We have three possibilities for the forecast error (e_t)

It can be pulsative, zero or negative

- If $A_t > F_t$ e_t is positives
- If $A_t < F_t$ e_t is negative

- If $A_t = F_t$ it is zero

Sources and measurement of error

There are a number of reasons (sources) of error. The following are possible sources of error:

1. The model may be inadequate due to:
 - The omission of an important variable
 - A change or a shift in the variable that the model can't deal with
 - The appearance of new variables (e.g. New competition)
2. Irregular variations/ Randomness may occur due to
 - Severe weather condition/ natural phenomena.
 - Temporary shortage.
3. The forecasting techniques may be used incorrectly or the result may be misinterpreted.

Measurement of errors

Four commonly used measures for summarizing forecasting errors are :

1. Mean absolute deviations (MAD).
2. Mean square error (MSE)
3. Mean absolute percent error (MAPE)
4. Tracking signal (TS)

1. Mean absolute deviations

MAD is average forecasting error based up on the absolute difference between the actual and the forecasted demand.

$$MAD = \sum_{i=1}^n |A_t - F_t| / n$$

Where n = total number of periods

A_t = actual demand for period t

F_t = forecasted demand for period t

$|$ = absolute symbol

MDA = average absolute error

2. Mean square error (MSE)

MSE is the average of squared errors

$$MSE = \sqrt{\frac{\sum (A_t - F_t)^2}{n - 1}}$$

3. Mean Absolute percent error (MAPE)

MAPE is the average absolute difference between the actual and forecasted demands expressed as a percentage of the actual demand. MAPE is a relative expression method as opposed to MAD and MSE which are absolute measures. MAPE is calculated by using the following formula.

$$MAPE = \frac{\sum_{i=1}^n \frac{|A_t - F_t|}{A_t}}{n}$$

4. Tracking Signal

Tracking signal is the ratio of cumulative forecasting error to the corresponding value of MAD, used to monitor forecast. The interest is to detect any bias in error overtime (i.e. tendency for a sequence of errors to be positive or negative).

The tracking signal is computed period by period by using the formula:

$$TS_t = \frac{\sum (A_t - F_t)}{MAD_t}$$

Example1. A specific forecasting model was used to forecast demands for a product. The forecasts and the corresponding demands that subsequently occurred are shown bellow.

Month	Actual	Forecast
Oct...	700	660
Nov...	760	840
Dec...	780	750
Jan...	790	835
Feb...	850	910
Mar...	950	890

Required

Calculate MAD, MES, MAPE and TSt